Resistive switching and memory computing

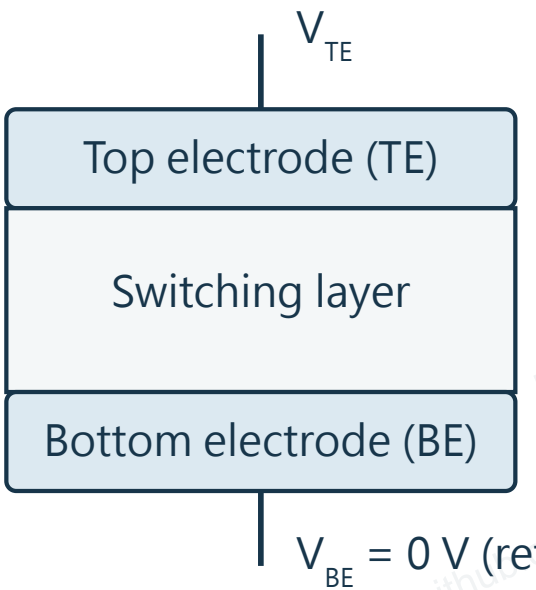
A resistive memory cell typically has a top electrode, switching layer and bottom electrode. A programming pulse changes its stored state; a specified read bias probes its current.

LRS and HRS mean low and high resistance. Assigning either state to a binary value is a system convention, not a material law. We connect measurements to physical models, arrays and reliability without treating one observation as proof of a unique mechanism.

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Cell structure and voltage reference

Two-terminal cell: define voltage first



Define $V = V_{TE} - V_{BE}$

Operations and readout

SET: higher conductance

RESET: lower conductance

Read: state voltage and duration

$G_{read} = I_{read} / V_{read}$

Polarity and current limit are device-specific; no particular mechanism is assumed.

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Read conductance has measurement conditions

At a specified voltage and time, define $G_{read} = I_{read}/V_{read}$. This ratio is an apparent conductance at that operating point.

Nonlinear response, polarity asymmetry or drift can make it different at another bias or time. A read can also disturb the stored state.

A voltage sweep traces a response under a specified sweep rate and current limit. Pulse experiments specify amplitude, duration and sequence and can probe switching and read disturbance.

Report temperature, voltage polarity, compliance, pulse timing and read protocol. A single curve does not establish device-to-device or cycle-to-cycle distributions.

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Forming, SET, RESET and read

- Forming is an initial conditioning step required by some devices, not all.
- SET generally moves to a more conductive state; RESET moves to a less conductive state.
- Read probes the state. Current compliance constrains programming current but does not guarantee freedom from damage.

For bipolar switching, SET and RESET generally use opposite bias polarities. Unipolar switching can use the same polarity with different pulse or current-limit conditions.

Define voltage explicitly, for example $V = V_{top} - V_{bottom}$, before naming positive or negative. A polarity observed in one stack is not universal.

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Trace a pulse protocol

This synthetic example assumes a particular bipolar stack and $V = V_{top} - V_{bottom}$; the numbers are illustrative, not specifications. Record the read immediately after each pulse and again later to detect drift.

Step	Applied pulse	Current limit	Read at 0.1 V	Inference
Initial read	+0.1 V, 10 µs	read circuit	1 µA	$G_{read} = 10$ µS
SET	+1.0 V, 1 µs	100 µA	20 µA	$G_{read} = 200$ µS
RESET	−1.0 V, 1 µs	not specified	1 µA	$G_{read} = 10$ µS

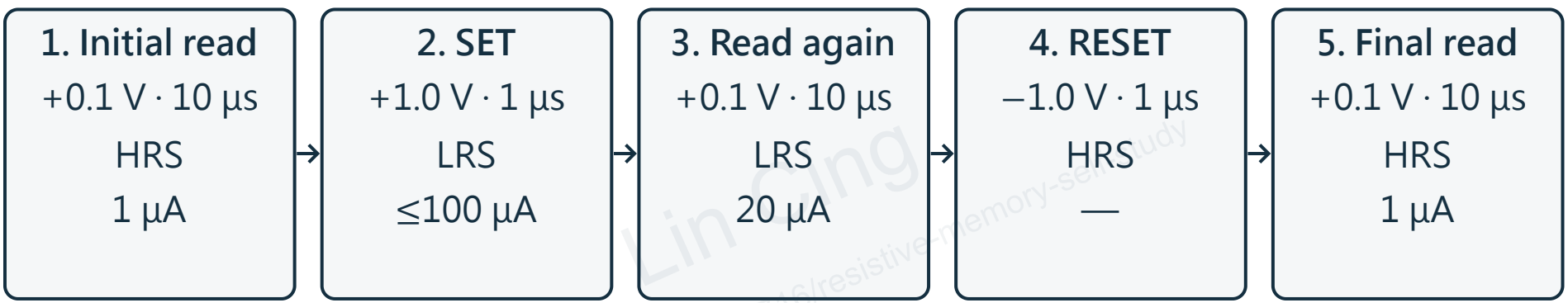
Compliance constrains programming current; a low read pulse is not automatically nondestructive. Verify repeated reads and record drift instead of inferring it from one measurement.

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Pulse operations and state sequence

Pulse and read sequence: illustrative example

Example only; V = Vtop − Vbottom.



Read currents shown; SET limited; RESET current unspecified.

Not to scale; reads assumed nondisturbing.

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ECM: a possible redox pathway

In electrochemical metallization (ECM), an active electrode can oxidize; metal cations migrate through the switching layer and reduction can form a conductive bridge. Subsequent operation can narrow or rupture it.

For a hypothetical Ag / oxide / inert-electrode stack with Ag positive during SET, oxidation $\mathrm{Ag} \rightarrow \mathrm{Ag}^{+} + e^{-}$ supplies cations while electrons enter the external circuit. Reduction $\mathrm{Ag}^{+} + e^{-} \rightarrow \mathrm{Ag}$ can deposit metal where electrons are available.

Nucleation, growth direction and RESET depend on electron supply, transport and reaction kinetics. This polarity and schematic are examples, not universal rules.

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ECM stages for one electrode configuration

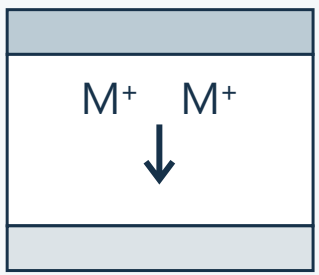
ECM: ions and a conducting bridge

Example: active top metal M (Ag or Cu), inert bottom; V = Vtop – Vbottom

Top: active electrode M | Middle: solid electrolyte | Bottom: inert electrode

① Oxidation

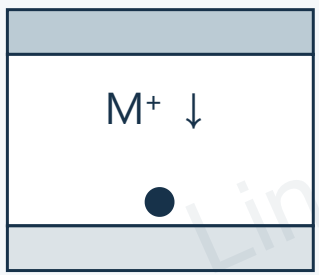
SET: V > 0



$M \rightarrow M^+ + e^-$

② Move + reduce

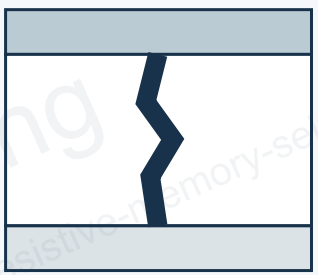
SET: V > 0



$M^+ + e^- \rightarrow M$

③ Bridge formation

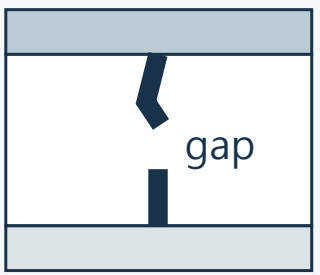
SET: compliance



May be LRS

④ Local rupture

Example: V < 0



May be HRS

Schematic, not to scale; growth direction depends on materials and reaction kinetics.

RESET polarity, rupture location and pulse depend on device; specify read conditions.

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VCM and competing explanations

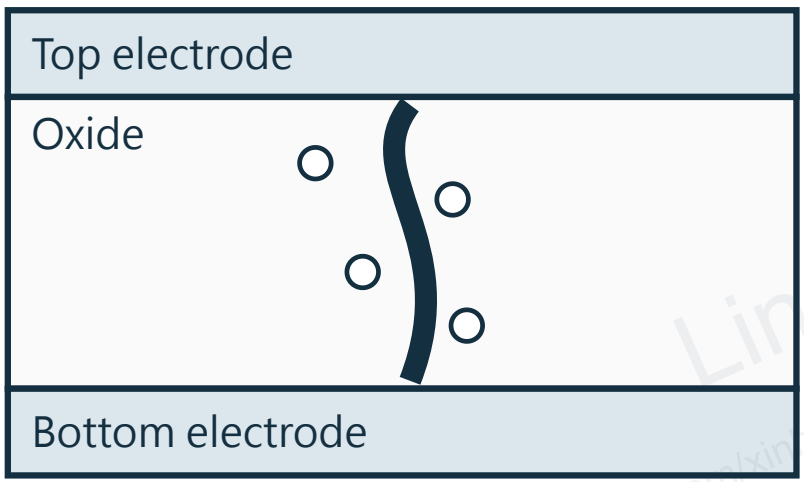
Valence-change mechanisms (VCM) involve oxygen-ion or oxygen-vacancy transport and local redox changes. Redistribution changes local stoichiometry and electronic transport. One model emphasizes a filament; another emphasizes an interface barrier. Each requires specific electrode, field and redox assumptions. Thermochemical contributions and unipolar behavior are not universally equivalent.

A hysteretic I–V curve is not proof of a unique mechanism. Local chemical maps, conductive microscopy, in-situ imaging and time-resolved measurements can test models, subject to resolution and possible measurement perturbations; correlation alone does not establish causality.

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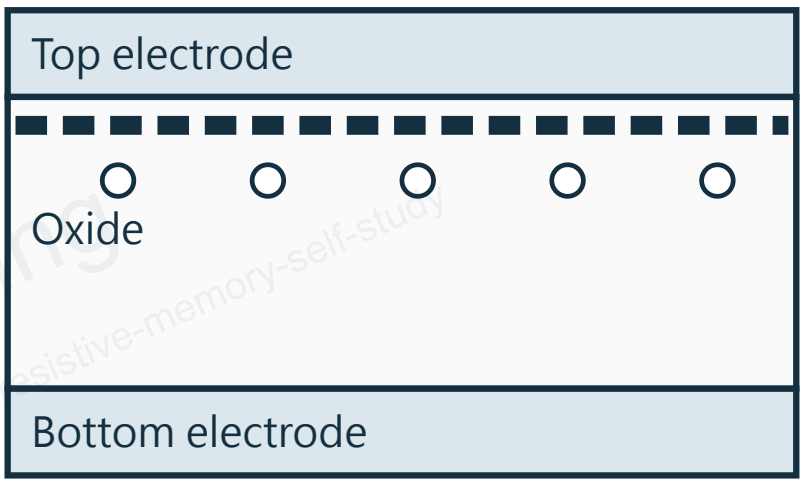
VCM: two possible models

Localized filament model



Local pathway changes

Interface-barrier model



Interface barrier changes

○: schematic oxygen defects | Models, not proof; defect charge depends on material and conditions.

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Model a state without assuming a mechanism

One framework is $i(t) = G(x(t), v(t), t)v(t)$ and $\dot{x} = f(x, v, t)$. The chosen state x may be multidimensional, and G need not be independent of voltage or time.

An ideal memristor is not synonymous with every two-terminal device displaying hysteresis. A model must state its domain and be tested against measurements.

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An ideal crossbar computes column currents

Let row i be driven at V_i and the conductance at row i, column j be G_{ij}. Assume linear read response, negligible wire resistance, and columns held at virtual ground.

Each branch carries $I_{ij} = G_{ij} V_i$. Kirchhoff's current law then gives $I_j = \sum_i G_{ij} V_i$. These assumptions are part of the result.

For $G = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$ $\mu\mathrm{S}$ and $V = (0.2, 0.1)\ \mathrm{V}$, the two column currents are $I_1 = 2(0.2) + 1(0.1) = 0.5\,\mu\mathrm{A}$ and $I_2 = 1(0.2) + 3(0.1) = 0.5\,\mu\mathrm{A}$. Here $\mu\mathrm{S} \cdot \mathrm{V} = \mu\mathrm{A}$; the numbers illustrate the model.

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Derive the array sum at the cell terminals

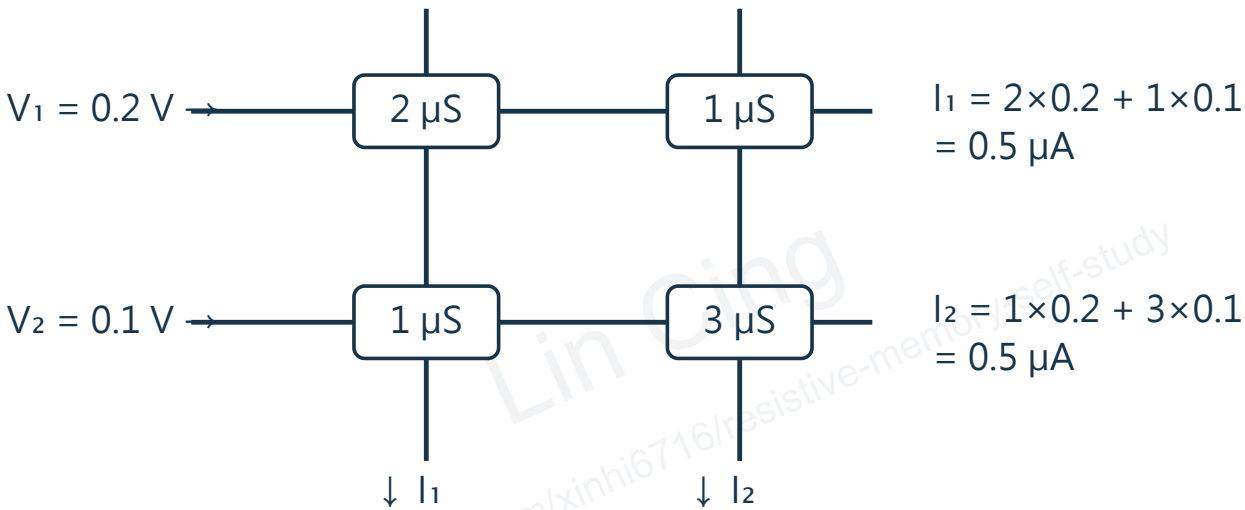
Let the local row and column potentials at cell (i, j) be v_{ij} and u_{ij}. For a linear read conductance, the branch current into the column is $I_{ij} = G_{ij}(v_{ij} - u_{ij})$. Kirchhoff's current law sums these branch currents at the column sensing node. Only when row wires are ideal and the column is held at virtual ground do $v_{ij} = V_i$ and $u_{ij} = 0$ yield $I_j = \sum_i G_{ij} V_i$.

With the earlier 2×2 example, suppose both local row-2 potentials are 0.08 V, row 1 remains 0.2 V, and column potentials remain zero. Then $I_1 = 2(0.2) + 1(0.08) = 0.48$ µA and $I_2 = 1(0.2) + 3(0.08) = 0.44$ µA. This assumes the local voltages are known; solving a real resistive network determines them rather than prescribing them.

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Ideal crossbar: two input rows

2 × 2 crossbar: sum branch currents by column



Assumptions: linear read, no wire resistance, virtual-ground columns.
$I_j = \Sigma_i G_{ij} V_i$; units: μS × V = μA.

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When the array differs from the model

Wire resistance changes local voltages; finite sensing impedance changes column potential. Leakage, unintended paths, nonlinear response, conductance drift and programming error can change the measured sum.

A negative weight needs an explicit representation, such as a difference of conductances, and corresponding sensing circuitry. A single passive conductance is not a negative weight.

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Reliability is a measurement, not a slogan

Retention describes distinguishability after a stated time and temperature, with a defined read and failure criterion. Endurance is a cycle count under stated waveforms and failure criterion.

Separate cycle-to-cycle from device-to-device variation. Read margin depends on the sensing circuit and distributions, not merely two nominal resistance values.

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Compare reliability only under matching protocols

Consider two hypothetical measurements, each on 100 cells. Test A reports 95 cells retaining distinguishable read currents after 1 hour at 85 °C; test B reports 98 cells after 1 hour at 25 °C. Both must specify read voltage and time, current thresholds, prior cycling and whether failed cells were excluded. Even with those records, different temperatures prevent treating the observed percentages as a controlled comparison of two materials.

For endurance, record the SET/RESET amplitudes, pulse widths, verify procedure and failure criterion. A short high-temperature retention measurement alone cannot establish 10-year retention at room temperature; an extrapolation requires a supported model, uncertainty and appropriate validation.

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Applications and device requirements

Four uses: operations and device demands

Storage	Write state → read and decide Focus: read margin, retention, write endurance
Stateful logic	Read state → update state Focus: operations may consume writes
Offline inference	Program weights → repeated reads Focus: read accuracy, drift and converters
Online training	Read and compute → update weights Focus: update error, asymmetry and endurance

Conceptual flows; operations and peripheral costs depend on architecture.

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Storage and selectors: the array sets the constraints

For digital storage, a read circuit must distinguish the programmed states at specified voltages and times. A large single-cell resistance ratio does not by itself guarantee a usable array read margin: wire resistance, sensing noise and currents through unselected cells also matter.

An access device can suppress unselected-path currents using gate control, as in 1T1R, or using a nonlinear two-terminal selector. Its leakage, drive current, switching speed and integration cost must be assessed together with the memory element; a selector is not automatically an ideal switch.

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Stateful logic can require repeated writes

In stateful logic, a programmed device state participates in a logic operation and the result is stored by a subsequent state change. The precise voltage sequence and number of writes depend on the circuit and its logic convention. A reconfigurable circuit instead may write a configuration once and use it for many operations; do not assume each operation rewrites its configuration.

Suppose the **same cell** receives exactly k writes per operation, operations occur at N per second, and this rate is constant for T seconds. The cell accumulates NkT writes. For $k = 2$, $N = 10^3\,\mathrm{s}^{-1}$ and $T = 60\,\mathrm{s}$, this is 1.2×10^5 writes. If writes are spread across cells, sum each cell's actual writes instead. A write is not necessarily a complete SET/RESET cycle; endurance depends on the specified pulses and failure criterion.

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Inference and training stress different properties

An analog crossbar can use programmed conductances to perform an approximate weighted sum. Offline inference programs weights before use: read noise, conductance drift, line resistance and ADC/DAC precision affect the result. Weight signs need an explicit representation, such as a differential pair.

Online training also updates conductances during learning. Update nonlinearity, asymmetry, quantization, stochasticity and write endurance can affect the training trajectory. Impact depends on the model, workload, mapping and compensation; neither inference nor training is automatically tolerant of every device imperfection.

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Temporal randomness differs from a device fingerprint

A stochastic switching device may produce different outcomes across repeated identical pulses. A random-number generator uses temporal variation; bias, serial correlation, readout and any post-processing must be characterized. Repeated switching also consumes endurance.

A physical unclonable function (PUF) instead aims to extract a reproducible device-specific response from spatial differences among devices. Its usefulness depends on stability under environmental changes, uniqueness across devices and measurement noise. Temporal randomness and spatial variation are different properties, even if a single device technology exhibits both.

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Phase-change memory: the thermal pathway

In a typical phase-change memory (PCM), a heater raises a chalcogenide region's temperature. A sufficiently long heating interval can allow crystal nucleation and growth, producing a more conductive SET state. A RESET pulse heats an active region above its melting range; rapid cooling can freeze a more resistive amorphous region.

These are illustrative thermal routes, not universal pulse recipes: nucleation and growth rates, actual temperature distribution, quench rate and phase history matter. Crossing a nominal temperature alone is not a guarantee of phase change. ECM and VCM can involve redox and ion transport as well as heating; a similar electrical hysteresis does not make their microscopic processes identical to PCM.

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A lumped thermal model

Let T_0 be ambient temperature, C_{th} thermal capacitance, R_{th} thermal resistance and P constant power during a pulse. The energy balance is $C_{th}\,dT/dt = P - (T - T_0)/R_{th}$, with $T(0) = T_0$. Its solution is $T(t) = T_0 + PR_{th}(1 - e^{-t/\tau})$, where $\tau = R_{th}C_{th}$.

This model assumes one uniform temperature, constant parameters and a known power waveform. A real pulse need not deliver constant power: resistance changes during switching and spatial hot spots can invalidate the single-node approximation.

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PCM pulse example: calculate before naming a state

Assume $T_0 = 300$ K, $R_{th} = 10^5$ K/W and $\tau = 10$ ns. A 4 mW pulse for 30 ns gives $T = 300 + 400(1 - e^{-3}) = 680.1$ K. In an independent trial starting at 300 K, a 10 mW pulse for 10 ns gives $T = 300 + 1000(1 - e^{-1}) = 932.1$ K.

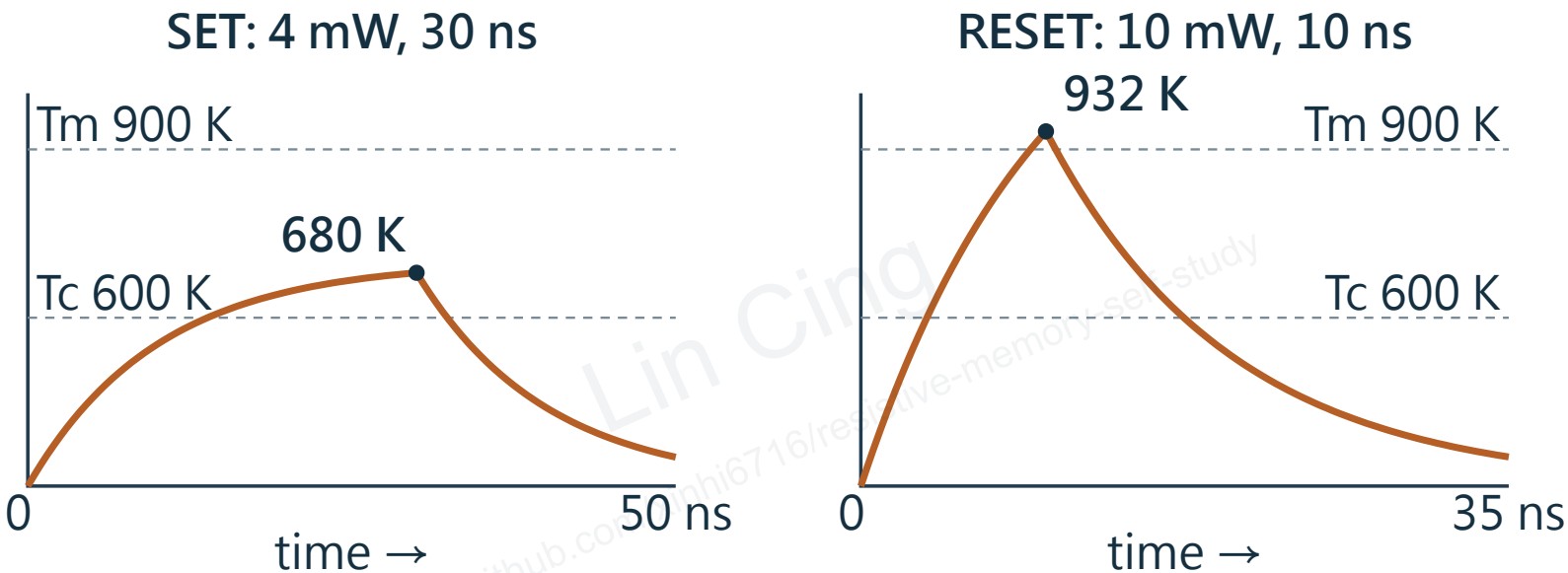
If illustrative crystallization and melting thresholds are 600 K and 900 K, respectively, the first peak lies between them and the second exceeds the melting threshold. These peaks alone do **not** prove SET or RESET: kinetics and cooling remain necessary to assess.

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Heating and quenching are separate stages

PCM thermal pulses: a lumped example

T_0 = 300 K; Rth = 10^5 K/W; τ = 10 ns



Assumed thresholds; temperature alone does not prove a phase change.

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Cooling after a RESET pulse

For the 932.1 K endpoint, set $P = 0$ after the pulse and keep T_0 and τ constant. Cooling follows $T(t) = 300 + (932.1 - 300)e^{-t/(10\,\mathrm{ns})}$; reaching 600 K takes $10\ln(632.1/300) = 7.45$ ns. This is a modelled cooling time, not proof that a particular quench rate yields a stable amorphous state.

The initial cooling rate is $dT/dt = -(932.1 - 300)/(10\,\mathrm{ns}) = -63.21$ K/ns. It slows as temperature approaches the ambient value. A spatial temperature profile or a change in thermal parameters can alter both the cooling time and the locally realized phase.

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A read-and-write OR: make the circuit boundary explicit

Consider two memory cells connected in parallel to a 0.1 V read supply and a virtual-ground current sensor. A stored 0 is $1\ \mu\mathrm{S}$, and a stored 1 is $10\ \mu\mathrm{S}$. Assume linear, stable, non-disturbing reads, ideal wiring and negligible sensor error. Then $I = 0.1(G_A + G_B)$ in μA.

A	B	Sum current (µA)	$I > 0.6\ \mu$A?
0	0	0.2	0
0	1	1.1	1
1	0	1.1	1
1	1	2.0	1

The comparator computes OR under the stated convention; it does not itself write a memory cell.

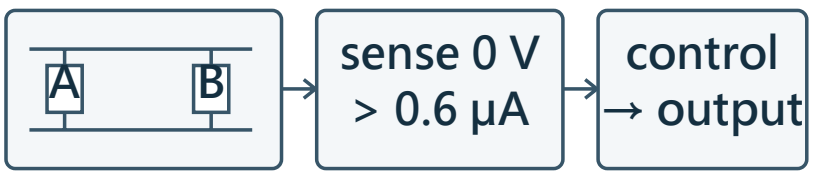
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Read-and-write OR and spike timing

Parallel OR read and synaptic pulses

1. Read, compare, then write

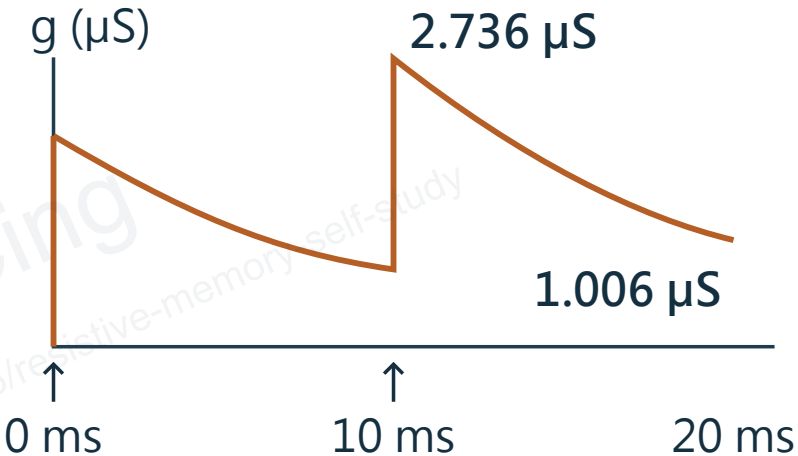
V = 0.1 V; 0/1 = 1/10 μS



A	B	I (μA)	OR
0	0	0.2	0
0	1	1.1	1
1	0	1.1	1
1	1	2.0	1

2. Two synaptic pulses

Δg = 2 μS; τ = 10 ms



Ideal pulses; exponential decay.

Hypothetical model; control writes a separate preset output.

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From the comparator to a stored result

To store the OR result, first initialize a separate output cell to HRS. The digital controller applies the appropriate SET pulse only for comparator output 1 and verifies the result. This example uses sensing and digital control; it is not evidence of an autonomous native stateful gate. Cell variation, sensing margin, leakage, read disturbance and write failures can change the outcome.

For the assumed currents, the low-state current is 0.2 µA and the smallest high-state current is 1.1 µA. With an ideal threshold of 0.6 µA, their distances from the threshold are 0.4 and 0.5 µA. These are nominal margins, not an error probability; noise and distributions must be measured.

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A dynamic synapse: events and decay

As an illustrative short-term model, between spikes $dg/dt = -g/\tau$ and each instantaneous spike adds Δg. This is a chosen dynamical model, not a claim that every resistive device follows exponential decay. Let $g(0^-) = 0$, $\tau = 10$ ms and $\Delta g = 2\,\mu$S, with spikes at 0 and 10 ms.

After the first spike $g(0^+) = 2\,\mu$S. Just before the second, $g(10^-) = 2e^{-1} = 0.7358\,\mu$S; just after it, $g(10^+) = 2(1 + e^{-1}) = 2.7358\,\mu$S. At 20 ms without another spike, $g(20) = 2.7358e^{-1} = 1.0064\,\mu$S. At a non-disturbing 0.1 V read, $I(10^+) = 0.27358\,\mu$A.

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Routing spikes is part of the computation

An event can carry a source address and timestamp through local and inter-core routers; a routing table maps the event to destination synapses. Multicast sends one event to several destinations, while programmed connectivity decides which synapses receive it. The dynamic update above must be evaluated at the actual arrival time.

For example, an event generated at 0 ms and delayed by 2 ms reaches a synapse at 2 ms; the next arrives at 12 ms if it has the same delay. Their separation remains 10 ms. Unequal delays, packet loss and congestion can change the effective spacing, so circuit behavior cannot be inferred from the source timestamps alone.

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Random switching: a probability model and its limits

If a specified experiment is modelled by a constant switching hazard λ, the waiting time is exponential: $P(T \le t) = 1 - e^{-\lambda t}$. Thus $\lambda = 10^6\,\mathrm{s}^{-1}$ and $t = 1\,\mu$s give $P(T \le t) = 1 - e^{-1} \approx 0.632$. A constant hazard, reset protocol, voltage and temperature are assumptions to test, not universal device laws.

For raw independent, identically distributed bits with $P(1) = 0.6$, min-entropy is $H_{min} = -\log_2(0.6) = 0.737$ bit per raw bit. This one-bit statistic alone does not measure entropy rate if bits are correlated or the distribution changes over time.

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Debiasing is conditional on the source

Under the independent, identically distributed model, nonoverlapping pairs 01 and 10 each have probability $0.4 \times 0.6 = 0.24$. Map 01 to 0 and 10 to 1; discard 00 and 11. The acceptance probability is 0.48 per pair, giving 0.24 output bits per raw bit on average, with equal probabilities for accepted 0 and 1.

Correlation, drift or selection bias can break that guarantee. Characterize the source, dependence, operating conditions and post-processing; passing a finite collection of statistical tests does not by itself prove cryptographic unpredictability.

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Material implication uses the stored state

Material implication is $p \Rightarrow q = \neg p \vee q$. Unlike the preceding read/comparator/write OR example, an implication operation can conditionally update the stored state of Q in the same circuit used for its inputs; P is intended to remain unchanged. The result depends on the initial states and a circuit-specific switching window.

For a concrete model, define 0 as HRS ($1\,\mu$S) and 1 as LRS ($10\,\mu$S). Connect the lower terminals of P and Q to one node, then connect that node to ground through $G_L = 3\,\mu$S. Apply $V_P = 0.5$ V and $V_Q = 1$ V at their upper terminals for the specified pulse. These values are illustrative, not device specifications.

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The voltage division decides whether Q changes

Assume each device initially behaves as an ideal linear conductance G_P, G_Q. Kirchhoff's current law gives $v = (0.5G_P + 1.0G_Q)/(G_P + G_Q + 3)$ V. The device voltages are $V_P^{dev} = 0.5 - v$ and $V_Q^{dev} = 1 - v$. Conductances are in µS.

p	q initially	v (V)	V_Q^{dev} (V)	q after pulse
0	0	0.3000	0.7000	1
0	1	0.7500	0.2500	1
1	0	0.4286	0.5714	0
1	1	0.6522	0.3478	1

Assume an HRS Q SETs when its positive voltage is at least 0.65 V for this pulse, and otherwise retains its state. Hence only 00 changes; the resulting column is 1, 1, 0, 1, the truth table for $\neg p \vee q$.

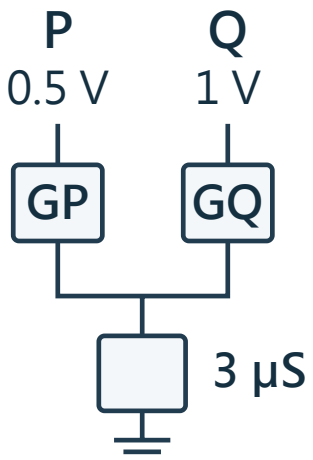
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Implication requires a verified switching window

IMPLY and stochastic AND

1. Conditional switching

0 = 1 μS; 1 = 10 μS



P Q	VQ (V)	Q out
0 0	0.700	1
0 1	0.250	1
1 0	0.571	0
1 1	0.348	1

VQ = 1 V − Vnode

Q: 0→1 if VQ ≥ 0.65 V

External initialization; P protected; Q assumed to latch.

2. Bitstream AND

Each time step: A ∧ B



Independent inputs:
0.6 × 0.5 = 0.30

If A = B = X:
P(X=1)=0.5 → 0.5

Not 0.5 × 0.5 = 0.25

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Check disturb and the post-switch state

In the four initial states the largest $|V_P^{dev}|$ is 0.25 V; suppose magnitudes below 0.3 V do not disturb P. For 00, once Q switches, the common node changes from 0.3000 to 0.7500 V. Assume Q latches in LRS even though its voltage then falls to 0.2500 V. These idealized switching, retention and non-disturbance conditions must be verified throughout real pulses, including variation and the intermediate circuit response.

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Stochastic multiplication needs independence

Let each pair of bits (A_i, B_i) represent probabilities $p = P(A_i = 1)$ and $q = P(B_i = 1)$; an AND gate outputs $Y_i = A_i B_i$. If A_i and B_i are independent at that index, $P(Y_i = 1) = pq$. With $p = 0.6$ and $q = 0.5$, the output mean is 0.3. For 100 independent identically distributed pairs, the count $S = \sum_{i=1}^{100} Y_i$ has $E[S] = 30$ and $\mathrm{sd}(S) = \sqrt{100(0.3)(0.7)} = 4.58$.

The cross-stream independence assumption matters: if $A_i = B_i = X_i$ for a fair bit X_i, each input still has mean 0.5 but $P(Y_i = 1) = 0.5$, not 0.25. Temporal correlations between pairs also alter the variance of S: $\mathrm{Var}(S) = \sum_i \mathrm{Var}(Y_i) + 2\sum_{i<j} \mathrm{Cov}(Y_i, Y_j)$.

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References and attribution

Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

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